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PROJECT REPORT

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CERTIFICATE

Certified that the project work and presentation entitled “AI Assistant for Pharmacist” is a bonafide work carried out by Abhilash Arun Jambagi (2SD22CS002), K K Sagar(2SD22CS039), Lalitkumar Choudhary(2SD22CS048) and Venkatesh Rao S (2SD23CS411), students of S. D. M. College of Engineering & Technology, Dharwad, in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belgaum, during the year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Project has been approved, as it satisfies the academic requirements in respect of project report prescribed for the said degree.

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ABSTRACT

*In the rapidly evolving healthcare and retail environment, traditional pharmacy management systems often rely on manual processes and fragmented digital tools, leading to inefficiencies, medicine wastage, billing errors, and limited data-driven decision-making. These challenges not only affect operational efficiency but also impact patient safety and sustainable resource utilization. This project, titled “**AI Pharmacy Management System**” addresses these limitations by integrating artificial intelligence with modern web technologies to create an intelligent, secure, and efficient pharmacy management solution.*

The primary objective of this project is to automate and optimize core pharmacy operations such as billing, stock management, invoice processing, and analytical reporting. The proposed system employs a modular architecture where a React-based frontend provides a responsive and user-friendly interface, while a Node.js and Express backend handles secure API communication, authentication, and business logic. PostgreSQL is used as the primary database to ensure reliable and structured data storage. A key methodological feature of the system is the integration of AI services for tasks such as invoice data extraction, intelligent analytics, and role-aware AI chatbot assistance, enabling faster decision-making and reduced manual intervention.

The system incorporates role-based access control to differentiate between administrative and worker-level functionalities, ensuring data security and controlled access to sensitive financial and inventory information. By leveraging AI-driven insights, real-time inventory tracking, and automated billing workflows, the project minimizes medicine wastage, reduces human errors, and enhances operational transparency. Through its intelligent design and sustainable digital practices, the AI Pharmacy Management System demonstrates how modern software solutions can improve healthcare service delivery while aligning with Sustainable Development Goals related to health, innovation, and responsible resource consumption.

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PROBLEM STATEMENT

Traditional pharmacy operations—such as sales recording, inventory management, stock updates from bills, expiry monitoring, and report generation—are still largely manual or only semi-automated. This creates several critical issues:

1. **High Manual Effort & Human Error:** Pharmacists and staff manually enter sales and stock data from printed or handwritten bills. Manual transcription and data entry lead to frequent mistakes (wrong medicine names, quantities, prices, expiry dates) which harm patient safety, inventory accuracy, and business accounting.
2. **Inefficient Inventory & Expiry Management:** Many pharmacies lack real-time visibility into stock levels and expiry status. Without automated monitoring and alerts, near-expiry or expired medicines can go unnoticed, causing financial losses, regulatory non-compliance, and health risk.
3. **Limited Decision Support & Slow Reporting:** Sales summaries, demand trends, and low-stock reports are often produced periodically or manually, delaying actionable insights. Owners and pharmacists cannot make fast, data-driven decisions for reordering, promotions, or compliance.
4. **Unfriendly User Interaction & Low Productivity:** Existing software typically relies on form-based interfaces and complex menus, which slow down routine tasks—especially for non-technical users—and reduce the ability to query the system quickly (e.g., “Which cough syrups expire next month?”).

This project addresses these problems by developing an **AI-powered Pharmacy Assistant** that combines advanced OCR (for handwritten bills) and Large Language Models (LLMs) to enable chat-based interaction, accurate automated stock updates, real-time inventory and expiry tracking, low-stock alerts, and on-demand report generation. Hence the problem title: “**AI-powered Pharmacy Assistant.**”

CHAPTER 1

INTRODUCTION

The **AI Assistant for Pharmacist** is a next-generation intelligent system engineered to modernize pharmacy operations by combining computer vision and large language models (LLMs). Built with contemporary web and AI technologies, the system delivers a conversational, chat-based interface that lets pharmacists and store owners perform routine tasks—such as sales recording, stock updates, expiry tracking, and report generation—using natural language or a quick photo of a bill. By automating data extraction from handwritten documents and providing on-demand analytics, the project aims to reduce manual effort, eliminate transcription errors, and enable faster, data-driven decisions in day-to-day pharmacy management.

1.1 Technological Foundation

The core of this project is built upon three pillars:

- **Frontend Interaction (React, Vite & Tailwind):** A responsive web interface offering a chat-like interaction model so users can query the system naturally (e.g., “Show me paracetamol expiring next month”) or upload bills/photos for instant processing. The UI is optimized for rapid workflows used in retail pharmacy counters.
- **Intelligent Language Layer (NLP & LLM APIs — Gemini / OpenAI):** An LLM-driven layer for intent detection, contextual understanding, entity normalization (mapping variant medicine names to canonical records), and conversational responses. It converts ambiguous or free-form inputs into precise database actions and human-readable reports.
- **Backend & Data Management (REST API + Relational DB):** A robust backend (e.g., Node.js/Express) connected to a relational database for sales, stock, and customer histories. It provides transactional integrity, real-time inventory calculations, analytics endpoints, and alerting logic.

1.2 Scope of the Project

The project focuses on automating pharmacy management using AI-driven chat interaction and photo-based stock updates. It enables real-time inventory and expiry tracking with automated alerts and report generation. The system reduces manual work and errors while supporting data-driven decision-making for small to medium pharmacies.

CHAPTER 2

SUSTAINABLE DEVELOPMENT GOALS (SDG)

The United Nations' 17 Sustainable Development Goals (SDGs) provide a comprehensive framework that increasingly shapes how engineering projects are conceived, designed, and executed worldwide. Engineers play a pivotal role in achieving these goals, as infrastructure, technology, and innovation are fundamental enablers of sustainable development. From SDG 6 (Clean Water and Sanitation) driving water treatment facility designs to SDG 7 (Affordable and Clean Energy) influencing renewable energy projects, these goals are no longer peripheral considerations but central to project planning and evaluation. Modern engineering projects are expected to demonstrate clear alignment with relevant SDGs, often requiring environmental impact assessments, social benefit analyses, and long-term sustainability metrics that extend beyond traditional cost-benefit calculations.

Engineering projects now commonly integrate multiple SDGs simultaneously, recognizing the interconnected nature of sustainable development. For instance, a transportation infrastructure project might address SDG 9 (Industry, Innovation and Infrastructure) while also contributing to SDG 11 (Sustainable Cities and Communities) through reduced emissions and improved urban mobility, and SDG 13 (Climate Action) through low-carbon construction methods. This multi-dimensional approach requires engineers to adopt systems thinking, considering not just technical performance but also social equity, environmental protection, and economic viability throughout a project's lifecycle.

The SDG framework has transformed engineering practice by establishing measurable targets and indicators that help quantify project impacts beyond traditional engineering metrics. Projects are increasingly evaluated on their contribution to goals like SDG 3 (Good Health and Well-being), SDG 12 (Responsible Consumption and Production), and SDG 15 (Life on Land), requiring engineers to collaborate with diverse stakeholders including environmental scientists, social planners, and community representatives. This paradigm shift has elevated the profession's responsibility, positioning engineers as key agents of sustainable development who must balance technical excellence with ethical considerations, resource efficiency, and long-term societal benefit in every project they undertake.

2.1 SDG Relevance of the AI Pharmacy Management System

The proposed **AI Pharmacy Management System** aligns with multiple Sustainable Development Goals by leveraging artificial intelligence and digital infrastructure to improve healthcare delivery and operational efficiency. The project strongly supports **SDG 3 (Good Health and Well-being)** by enabling accurate medicine management, timely stock updates, and intelligent billing and analytics, which reduce human error and improve patient safety. It contributes to **SDG 9 (Industry, Innovation and Infrastructure)** using AI-driven automation, secure backend services, and data-driven decision support that modernize pharmacy operations. Additionally, the system aligns with **SDG 12 (Responsible Consumption and Production)** by minimizing medicine wastage through precise inventory tracking, expiry monitoring, and demand-based stock insights generated by AI. By integrating AI analytics, secure data handling, and efficient resource utilization, the project demonstrates how digital health technologies can promote sustainable, reliable.

CHAPTER 3

OBJECTIVES

The specific technical and functional objectives of this project are as follows:

- 1 **To Develop a Chat-Based AI Pharmacy Assistant:** Design and implement an intelligent conversational interface that enables pharmacists and owners to interact with the system using natural language queries. The assistant will utilize NLP and LLMs to interpret user intent and provide instant responses for sales reports, stock availability, and customer-related information.
- 2 **To Automate Stock Entry Using AI and OCR:** Integrate Optical Character Recognition (OCR) with AI models to automatically extract medicine details from printed or handwritten purchase bills. This automation ensures accurate inventory updates, reduces manual data entry, and minimizes human errors.
- 3 **To Implement Smart Inventory Management with Alerts:** Develop an automated inventory monitoring system that tracks medicine quantities and expiry dates in real time. The system will generate low-stock and near-expiry alerts to ensure timely restocking and regulatory compliance.
- 4 **To Design a Responsive and User-Friendly Web Interface:** Build a clean, intuitive, and responsive web application that integrates all system features. The interface will be optimized for usability across devices, reducing the learning curve and improving operational efficiency for non-technical users.
- 5 **To Enable Secure Online Backup and Data Recovery:** Implement a cloud-based backup mechanism to securely synchronize sales, inventory, and customer data. This ensures data availability and continuity of operations in case of local system failures or hardware issues.

CHAPTER 4

LITERATURE REVIEW

I. Smart Pharmacy Management Systems

These studies focus on automating pharmacy operations using IoT, rule-based systems, and basic analytics to improve inventory monitoring and decision support.

1. Smart Pharmacy Management System Using AI and IoT (Reddy et al., 2021)

- **Achieved Outcomes:**
 - Automated monitoring of medicine stock levels and expiry dates.
 - Generated alerts for low stock and near-expiry medicines.
 - Assisted pharmacists in basic decision-making using analytics.
- **Key Features:**
 - IoT sensors for real-time inventory tracking.
 - Automated expiry alerts.
 - Basic AI-based trend analysis.
- **Limitations:**
 - Lacked advanced AI for intelligent analysis.
 - No chat-based or conversational interface.
 - Did not support prescription or bill image reading.

II. AI-Based Prescription and Document Recognition

These studies explore the use of OCR and machine learning techniques to digitize handwritten prescriptions and medical documents.

2. AI-Based Prescription Recognition System (Kumar & Singh, 2022)

- **Achieved Outcomes:**
 - Successfully digitized handwritten prescriptions.
 - Extracted medicine names, dosages, and quantities.
 - Reduced manual effort in recording prescriptions digitally.
- **Key Features:**
 - Utilized Tesseract OCR for text extraction.
 - Applied CNN models for handwriting recognition.
 - Focused on prescription digitization.
- **Limitations:**
 - Accuracy drops with poor handwriting quality.

- No integration with inventory or billing systems.
- Lacked NLP-based correction and contextual understanding

3. Optical Character Recognition in Medical Applications (Zhou et al., 2019)

- **Achieved Outcomes:**
 - Improved digitization of medical records and prescriptions.
 - Enhanced OCR accuracy through preprocessing techniques.
- **Key Features:**
 - Image preprocessing for noise reduction.
 - OCR-based text extraction from healthcare documents.
 - Focus on improving recognition accuracy.
- **Limitations:**
 - Highly dependent on handwriting quality.
 - Minimal NLP-based semantic correction.
 - No automation beyond text extraction.

III. Automated Medical Store Management Systems

These works aim to reduce manual bookkeeping using desktop or web-based pharmacy management software.

4. Automated Medical Store Management Using Python and MySQL (Sharma et al., 2023)

- **Achieved Outcomes:**
 - Automated billing, sales tracking, and report generation.
 - Reduced paperwork and manual bookkeeping.
- **Key Features:**
 - Desktop-based application architecture.
 - Python used for backend logic.
 - MySQL database for data storage.
- **Limitations:**
 - Required manual data entry for stock updates.
 - No OCR or AI-based automation.
 - Limited to local system access only.

IV. AI-Powered Healthcare and Pharmacy Analytics

These studies apply AI techniques to analyze healthcare and pharmacy data for prediction and optimization.

5. AI-Powered Healthcare Data Analytics System (Gupta et al., 2022)

- **Achieved Outcomes:**
 - Predicted medicine demand trends.
 - Optimized stock levels using historical data.
 - Provided dashboards for analytics and visualization.

- **Key Features:**
 - Machine learning models for demand forecasting.
 - Analytical dashboards for decision support.
 - Sales and patient data analysis.

- **Limitations:**
 - Focused only on analytics, not automation.
 - No prescription or bill processing.
 - Lacked chat-based interaction and real-time updates.

CHAPTER 5

DETAILED DESIGN

5.1 Use Case Diagram

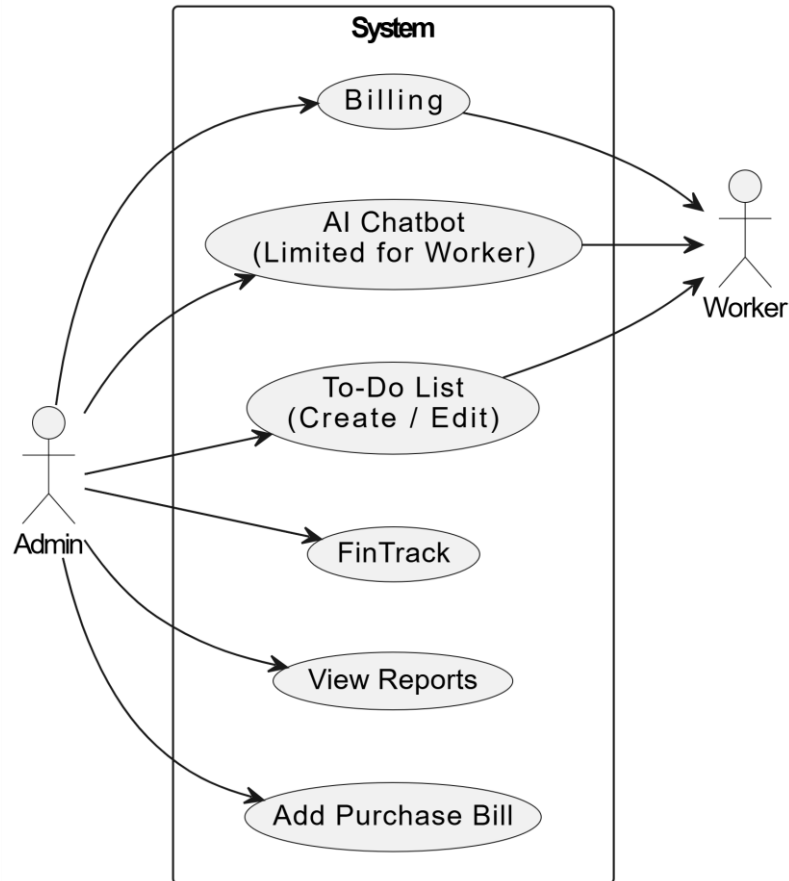


Figure 5.1: Use Case Diagram

Actors

- **Admin:** System administrator with full access and management privileges.
- **Worker:** System user with limited access to operational features.

Use Cases

1. Billing
 - Admin and Worker can create and manage billing entries.
 - View billing details related to purchases and transactions.
2. AI Chatbot
 - Admin can interact with the AI chatbot to access full system insights.
 - Worker can interact with the AI chatbot with **restricted access**, receiving only limited and role-appropriate information.
3. To-Do List Management

- Admin and Worker can create, update, and delete tasks.
 - Used to track daily work activities and responsibilities
4. FinTrack (Admin Only)
- Admin can monitor financial transactions and cash flow.
 - Used for tracking income, expenses, and financial performance.
5. Reports (Admin Only)
- Admin can generate and view analytical reports.
 - Reports provide insights into billing, financial records, and system usage.
6. Add Purchase Bill (Admin Only)
- Admin can add new purchase bills into the system.
 - Includes supplier details, bill amount, date, and payment status.

5.2 E-R Diagram:

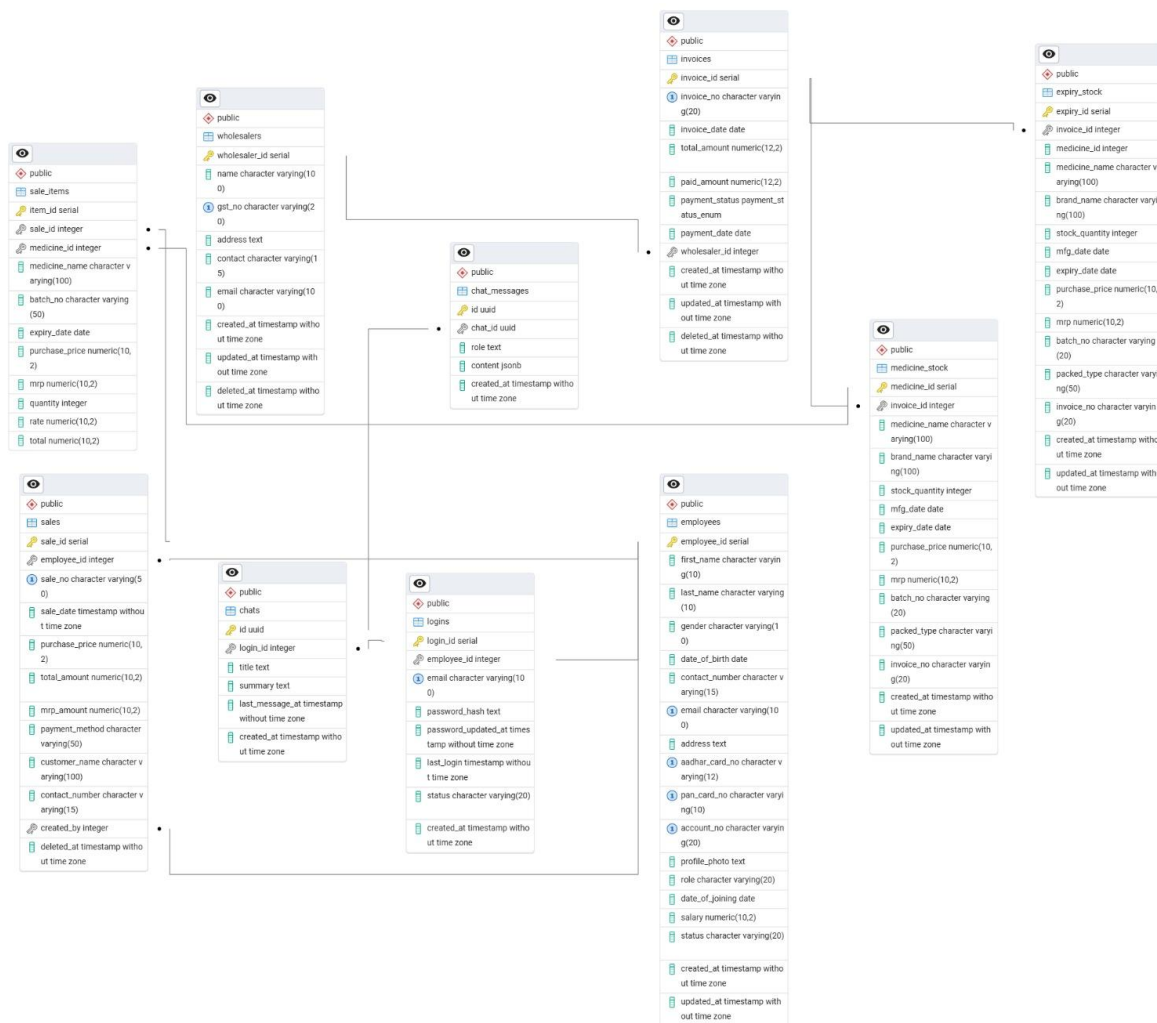


Figure 5.2: E-R Diagram

Database Schema Overview

The architecture is built on a relational model that ensures data integrity and minimizes redundancy. The system is divided into four primary modules:

- Inventory & Stock Management:** The core of the system revolves around the **medicine_stock** and **expiry_stock** entities. These tables track critical product data, including batch numbers, manufacturing/expiry dates, and real-time quantities. The separation of expiry_stock allows the system to isolate and manage non-sellable items efficiently.
- Procurement & Supply Chain:** The **wholesalers** table stores vendor information, while the **invoices** table records incoming shipments. Each invoice is linked to a wholesaler, providing a clear audit trail of where stock originated and the associated costs (purchase price vs. MRP).
- Sales & Transactions:** Customer transactions are handled through a master-detail relationship between **sales** and **sale_items**. The sales table captures the header information (date, total amount, customer contact), while sale_items records the specific medicines sold, their rates, and quantities, linked directly back to the medicine inventory.

- **User Management & Security:** The system employs a secure authentication layer. The **employees** table stores sensitive personal and professional data (Salary, PAN, Aadhar), which is linked to the **logins** table. This separation ensures that credentials (hashed passwords) are isolated from general employee profiles.
- **Communication Module:** A built-in messaging system is represented by **chats** and **chat_messages**. This facilitates internal coordination, allowing employees to communicate within the platform, with messages linked to specific chat threads for organized history.

Key Technical Relationships

The diagram utilizes **One-to-Many (1:N) relationships** extensively. For instance, one wholesaler can have multiple invoices, and one sale transaction can contain multiple sale items. This structure ensures that the database can scale as the volume of inventory and transactions grows.

5.3 Sequence Diagram:

1. Billing Sequence Diagram

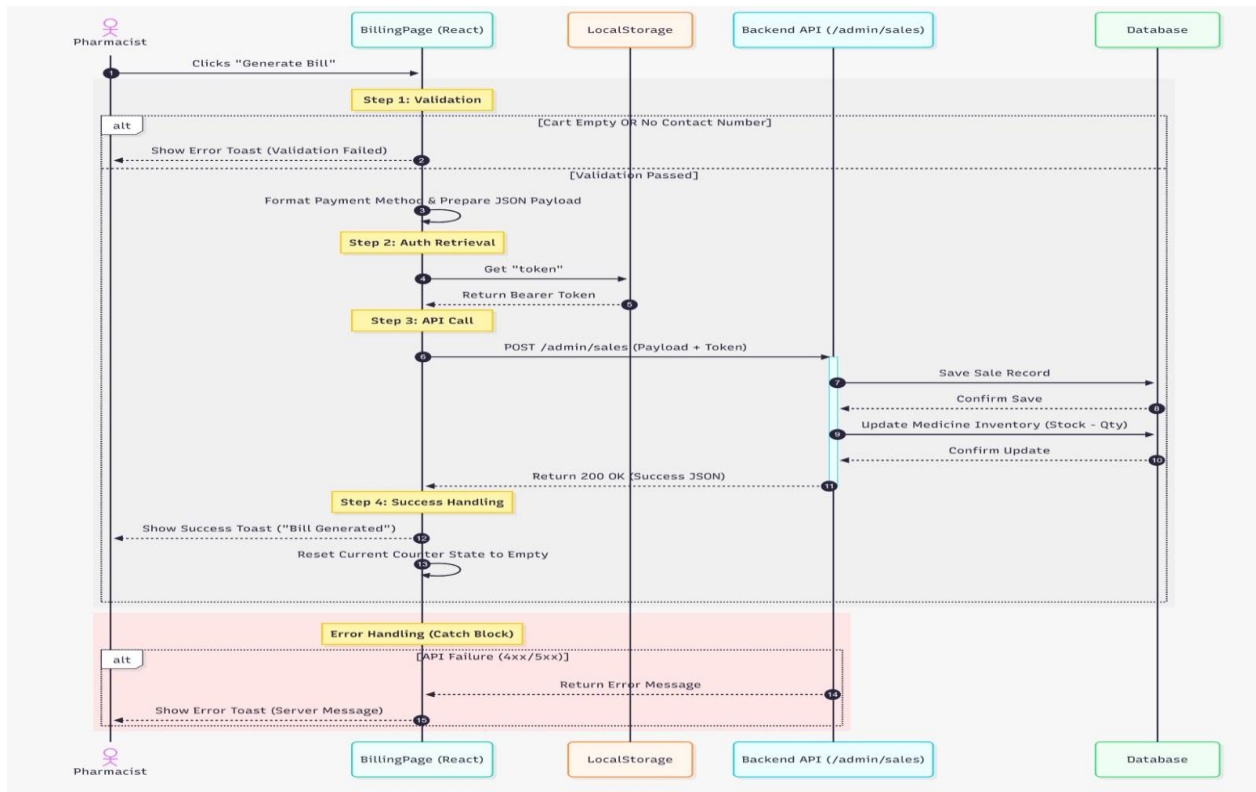


Figure 5.3.1: Billing Sequence Diagram

Sequence Diagram Description

The sequence diagram illustrates the complete interaction flow involved in generating a sales bill in the pharmacy management system. The workflow is initiated by the Pharmacist through the React-based billing interface and coordinated between the frontend, browser storage, backend API, and database. The process is divided into four critical phases to ensure data validation, secure processing, accurate inventory updates, and reliable error handling.

Steps Involved in the Sequence Diagram

- **Bill Generation Initiation:** The process begins when the pharmacist clicks the “Generate Bill” button on the Billing Page, triggering the billing workflow.
- **Client-Side Validation:** The system validates the cart details and customer contact information before allowing the process to continue.
- **Validation Failure Handling:** If validation fails, an error message is displayed to the pharmacist and the workflow stops without contacting the backend.
- **Payload Preparation:** After successful validation, the billing data and payment method are formatted into a structured JSON payload.
- **Authorization Token Retrieval:** The Billing Page retrieves a Bearer token from the browser’s LocalStorage to authenticate the API request.

- **API Request Submission:** The frontend sends a POST request to the backend sales API along with the billing payload and authorization token.
- **Sales Record Storage:** The backend processes the request and stores the sales transaction details in the database.
- **Inventory Update:** The system updates the medicine inventory by reducing stock quantities based on the items sold.
- **Database Confirmation:** The database confirms the successful insertion of the sales record and the inventory update.
- **Backend Response:** A successful response is returned to the frontend indicating that the billing operation has completed successfully.
- **Success Handling:** The Billing Page displays a success message to the pharmacist and resets the cart for the next transaction.
- **Error Handling (Alternate Flow):** If an error occurs during the API call, an error message is returned and displayed without clearing the cart.

2. Stock-Entry Sequence Diagram

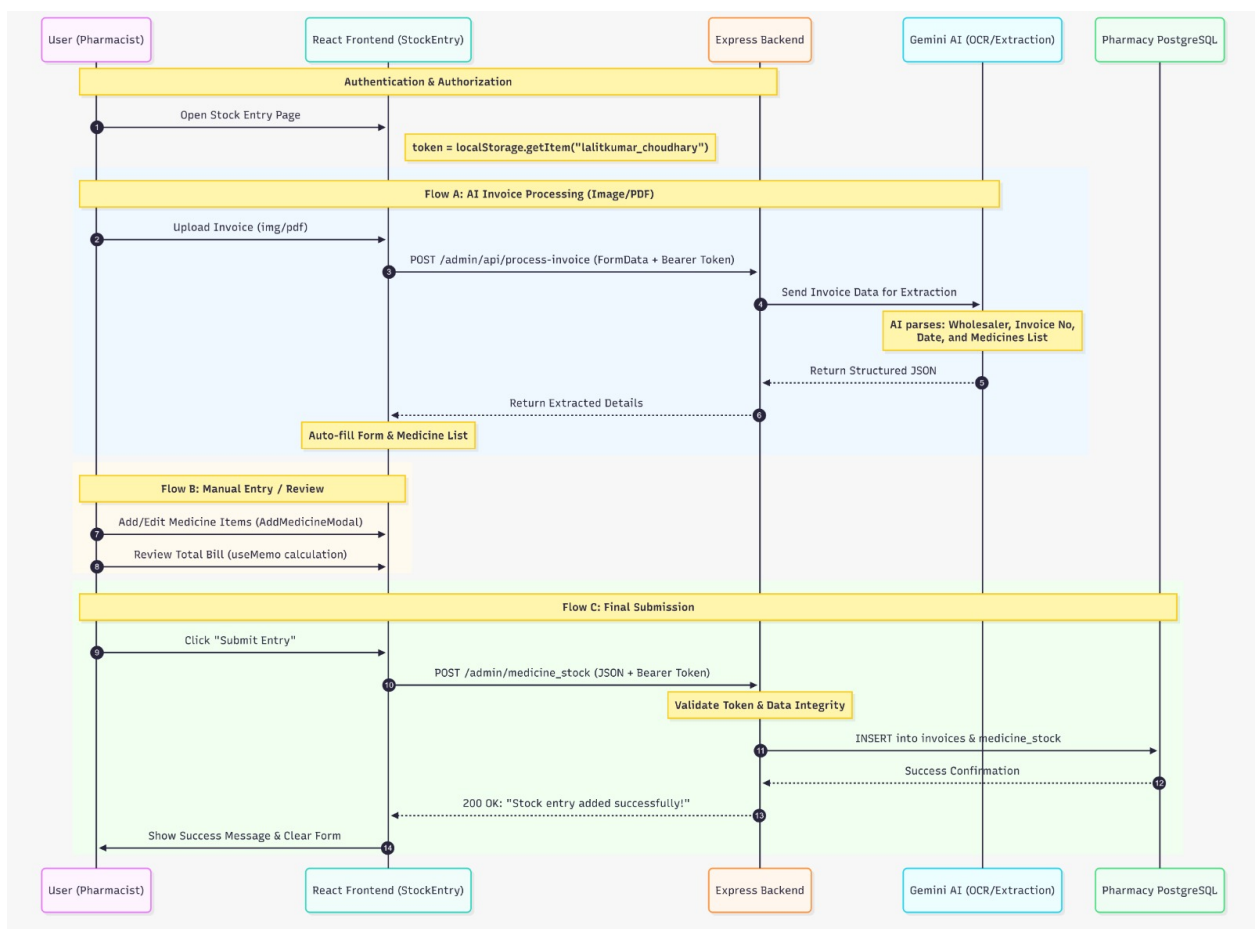


Figure 5.3.2 Stock-Entry Sequence Diagram

Sequence Diagram Description

The sequence diagram illustrates how a user interacts with the system to authenticate and request AI-powered analytics. It shows the coordinated flow between the frontend UI, backend services, AI orchestrator, and database to generate secure, role-aware insights.

Steps Involved in the Sequence Diagram

- **User Login Input:** The user enters login credentials in the frontend UI to initiate the authentication process.
- **Login Request to Backend:** The frontend sends a login request to the Express backend using the /api/login endpoint.
- **Credential Validation:** The backend validates the provided credentials against the Auth/Employee database.
- **User Profile Retrieval:** Upon successful validation, the database returns the user's role and employee ID to the backend.
- **Session Context Creation:** The backend sends an authentication success response, and the frontend stores the user context (role and ID).
- **Analytics Request Input:** The user submits an analytics query such as "Show my sales today" through the frontend UI.
- **Analytics API Call:** The frontend sends the analytics request to the backend along with user context and a request summary.
- **System Context Injection:** The backend enriches the request by injecting predefined system context and prompt configuration.
- **AI Orchestrator Processing:** The backend forwards the request to the Gemini Orchestrator, which performs validation, identity pinning, and SQL generation.
- **Clarification Handling (Conditional):** If the request is ambiguous, the system prepares a clarification response instead of executing a query.
- **SQL Query Execution:** When a valid SQL query is generated, a read-only SELECT query is executed on the Pharmacy PostgreSQL database.
- **Result Set Evaluation:** The backend checks whether the query returned data and applies a fallback response if the result set is empty.
- **Response Mapping:** The retrieved data is formatted into a structured backend response suitable for frontend rendering.
- **Final Response Delivery:** The backend sends the final JSON response containing data tables and AI insights to the frontend UI.
- **UI Rendering:** The frontend renders the analytics results and AI-generated insights for the user.

3. AI Assistance Sequence Diagram

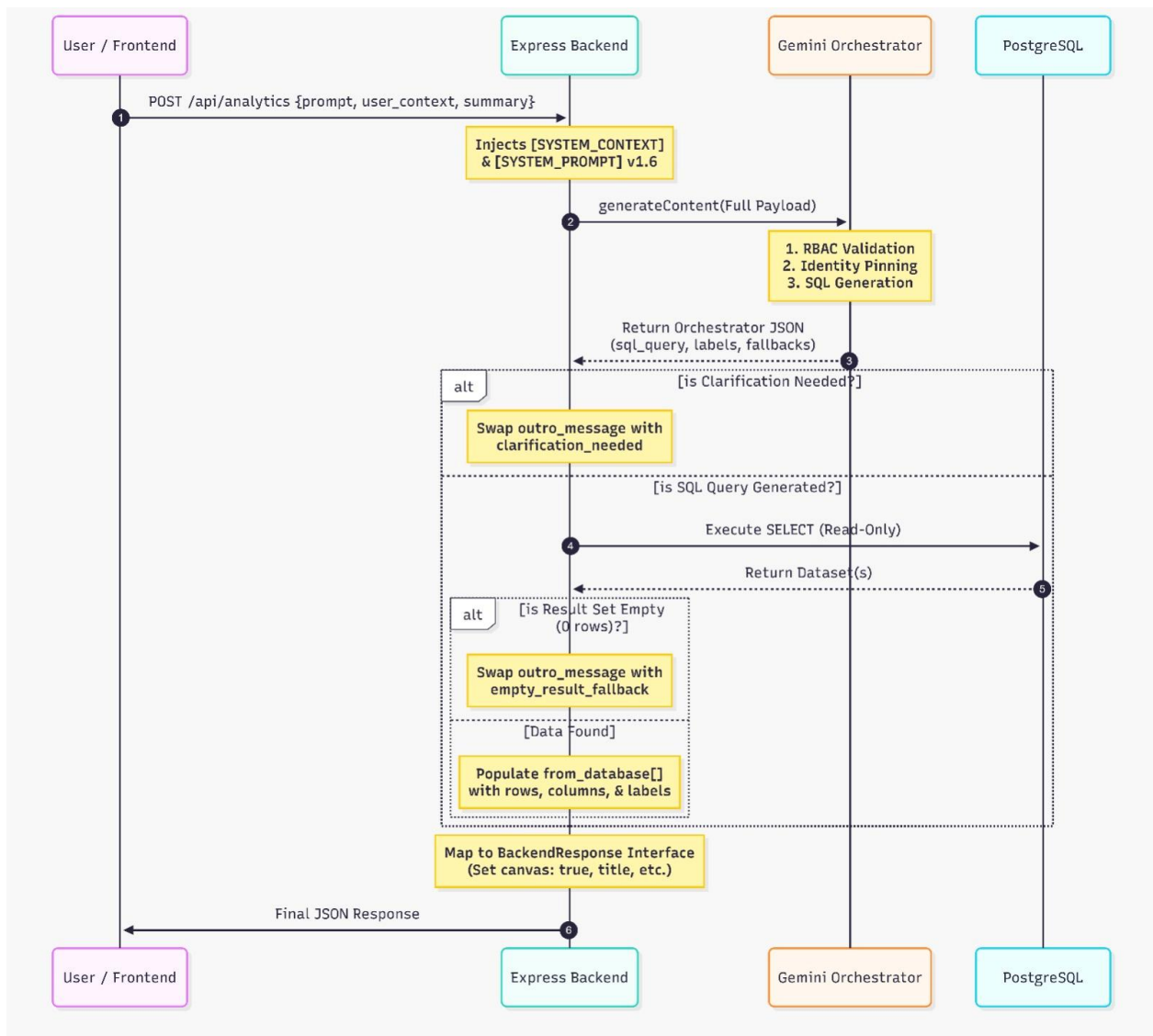


Figure 5.3.3 AI Assistance Sequence Diagram

Sequence Diagram Description

The sequence diagram represents the workflow for **stock entry management** in the pharmacy system. It illustrates how invoice data is processed using AI-based extraction, reviewed manually by the pharmacist, and finally stored securely in the database.

Steps Involved in the Sequence Diagram

- **Open Stock EntryPage:** The pharmacist opens the stock entry page in the React frontend, and the system retrieves the authentication token from local storage.
- **Invoice Upload:** The pharmacist uploads an invoice file in image or PDF format to begin automated invoice processing.
- **Invoice Processing Request:** The frontend sends the invoice file along with the Bearer token to the backend invoice processing API.
- **AI Extraction Request:** The backend forwards the invoice data to the Gemini AI service for OCR and information extraction.

- **Invoice Data Extraction:** The AI extracts structured information such as wholesaler name, invoice number, date, and medicine details.
- **Extracted Data Response:** The AI returns structured JSON data to the backend, which is then forwarded to the frontend.
- **Auto-Fill Stock Entry Form:** The frontend automatically fills the stock entry form and medicine list using the extracted invoice data.
- **Manual Review and Editing:** The pharmacist reviews the extracted details and manually adds or edits medicine items if required.
- **Bill Review:** The frontend calculates and displays the total bill amount for final verification.
- **Final Submission:** The pharmacist clicks the “Submit Entry” button to confirm and submit the stock entry.
- **Backend Validation:** The backend validates the authentication token and verifies the integrity of the submitted data.
- **Database Insertion:** The backend inserts the invoice and medicine stock details into the PostgreSQL database.
- **Success Response:** The database confirms the insertion, and the backend returns a success response to the frontend.
- **UI Success Handling:** The frontend displays a success message and clears the stock entry form for the next operation.

5.4 Activity Diagram: Sending a Message

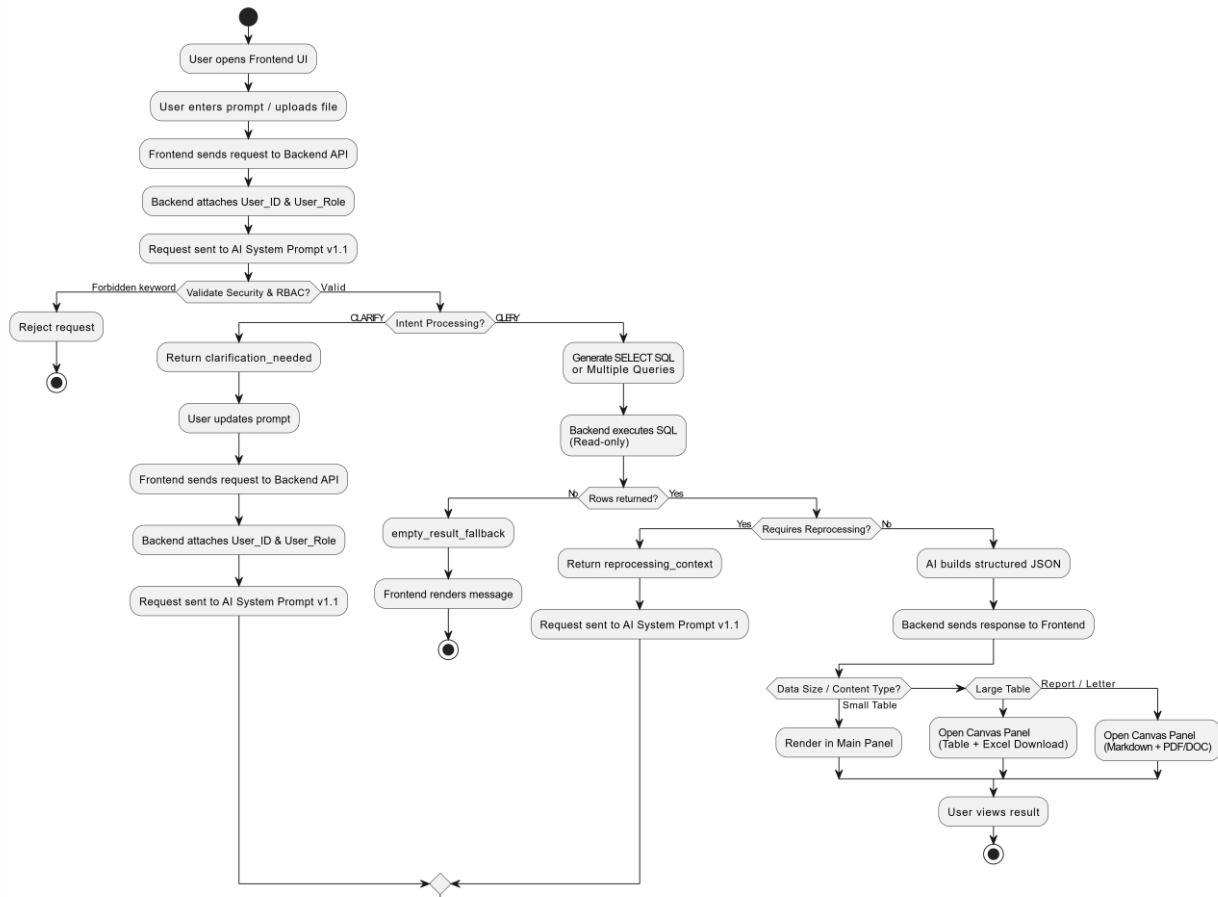


Figure 5.4 Activity Diagram

Activity Diagram Description

The activity diagram represents the end-to-end flow of a secure, role-aware AI query processing system. It illustrates how a user request is validated, interpreted by an AI system, processed through controlled database access, and finally rendered in the user interface based on the nature and size of the result. The diagram emphasizes security enforcement, clarification handling, and adaptive response generation.

- **Composition:** The process begins when the user opens the frontend interface and submits a prompt or uploads a file, initiating a request for AI-assisted data processing.
- **Request Mediation:** The frontend forwards the request to the backend, where the system attaches the user's identity and role information before sending the request to the AI system prompt.
- **Security & RBAC Validation:** The AI system validates the request against security rules and role-based access control policies. If a forbidden keyword or unauthorized intent is detected, the request is rejected immediately.
- **Intent Interpretation:** If the request is valid, the AI determines whether the intent requires clarification or direct query execution.
- **Clarification Handling:** When clarification is needed, the system returns a clarification request to the user, who updates the prompt and resubmits it through the same processing pipeline.
- **Query Execution:** For query-based intents, the AI generates read-only SQL queries that are executed by the backend against the database to fetch the required data.

- **Result Evaluation:** If no records are returned, an empty-result fallback message is generated and displayed. If data is returned, the system evaluates whether reprocessing is required for refinement.
- **Response Construction:** Once finalized, the AI constructs a structured JSON response that is sent back to the frontend by the backend.
- **Adaptive Rendering:** The frontend renders the result based on content type and size, displaying small datasets in the main panel and larger tables or reports in a canvas panel with export options.
- **Termination:** The activity concludes when the user views the final output or receives an appropriate fallback or error message.

CHAPTER 6

PROJECT SPECIFIC REQUIREMENTS

6.1 Hardware Requirements

- **Client Devices:** Any PC, laptop or smartphone with at least 4 GB RAM for day-to-day use.
- **Network:** Reliable internet connection for API calls, OCR uploads and cloud services (min. 512 Kbps recommended; higher bandwidth advised for frequent image uploads).

6.2 Software Requirements

- **Operating System:** Platform independent (Windows, macOS, Linux, Android, iOS).
- **Browser:** Modern web browser for the frontend (Chrome 80+, Firefox 75+, Safari 14+, Edge).
- **Frontend Stack:** HTML, CSS, JavaScript, React.js, Tailwind CSS.
- **Backend Stack:** Node.js (Express) or Flask (Python).
- **Database:** PostgreSQL or MongoDB for transactional and inventory data.
- **AI & OCR Tools:** OpenAI API (or equivalent LLM) for chat/NLP.
- **Version Control & Tools:** Git, GitHub; Visual Studio Code; Postman for API testing.
- **Hosting / Deployment:** Vercel, Render, AWS, or GCP as required.

6.3 Functional Requirements

- **Automated Stock Entry (OCR):** Uploading a bill/photo must trigger OCR + AI parsing to extract medicine name, quantity, batch/expiry and update inventory automatically.
- **Real-Time Inventory & Alerts:** Inventory levels and expiry statuses must update immediately after transactions or OCR imports; low-stock and near-expiry alerts must be generated automatically
- **Data Reliability & Backup:** Transactions and inventory changes must be synchronized to cloud backup (automatic sync/daily backup) to protect against local system failures.
- **Responsive UI:** Web interface must be responsive (Tailwind CSS) and usable across desktop and mobile viewports.

CHAPTER 7

IMPLEMENTATION

This section describes the high-level logic and architectural workflow of the AI Assistant for Pharmacist. The implementation follows a modular, cloud-assisted architecture, where the backend manages authentication, data processing, and persistence, while AI services handle intelligent automation such as OCR, NLP, and chat-based interactions.

7.1 Authentication and User Management Logic

The system functions as a secure access gateway to protect sensitive pharmacy data and ensure authorized usage.

- **User Authentication:**
Pharmacists and owners authenticate using secure credentials (email/username and password). Upon successful login, a user session is established, granting access based on role (admin/pharmacist).
- **Session Management:**
Authenticated sessions are maintained using secure tokens (e.g., JWT). These sessions allow continuous interaction with the system while preventing unauthorized access.

7.2 AI-Based Stock Entry and Data Processing

Since manual stock entry is error-prone, the system automates this process using AI and OCR technologies.

- **Image Upload:**
The user uploads a photo of a purchase bill or handwritten stock document through the web interface.
- **OCR Processing:**
The image is processed using Tesseract OCR to extract raw text, including medicine names, quantities, batch numbers, and expiry dates.
- **AI Interpretation:**
Extracted text is sent to an AI model (via OpenAI/Gemini API) for semantic understanding, correction, and normalization of medicine names.
- **Database Update:**
Validated data is cross-checked with existing inventory records and automatically stored in the database, updating stock levels in real time.

7.3 Chat-Based AI Interaction and Query Handling

Once data is stored, the system enables intelligent, conversational interaction for daily operations.

- **Natural Language Queries:**
Users interact with the system using chat queries such as “Show low-stock medicines” or “Generate today’s sales report.”.
- **Intent Detection:** The NLP engine identifies user intent (inventory check, sales query, report generation) and extracts relevant entities.
- **Response Generation**

The backend fetches required data from the database and formats the response using AI, presenting results in a human-readable conversational form.

- **Real-Time Insights**

Results are returned instantly, enabling faster decision-making without navigating complex interfaces

7.4 Inventory Monitoring, Alerts, and Backup

- **Real-Time Monitoring:**

Inventory levels and expiry dates are continuously monitored after each transaction or stock update.

- **Automated Alerts:**

The system generates alerts for low-stock and near-expiry medicines, helping prevent shortages and wastage.

- **Cloud Backup:**

All critical data (sales, inventory, customer records) is synchronized with secure cloud storage, ensuring data recovery in case of local system failure.

RESULTS

The project was successfully compiled and deployed. The following results were observed:

8.1 Login Page:

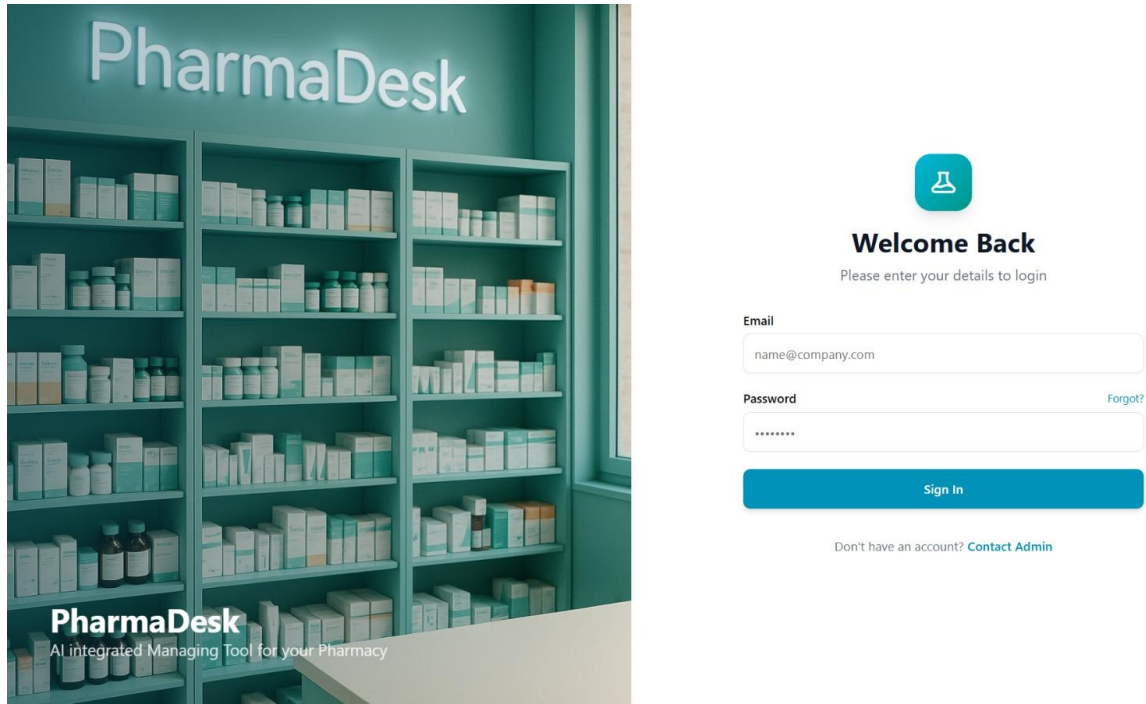


Figure 8.1 Login Page Image

The **User Authentication Module** acts as the secure gateway to the pharmacy system, employing a minimalist and professional interface to ensure ease of use. It utilizes **JSON Web Tokens (JWT)** to manage encrypted sessions, providing a high level of security by verifying every server request. The module is integrated with **Role-Based Access Control (RBAC)**, which identifies the user's credentials to dynamically restrict or grant access to sensitive areas, such as the AI panel or financial dashboard. Built with **CORS** protocols and secure error handling, the page maintains system integrity while remaining fully responsive for access across both desktop and mobile devices within the same network.

8.2 Dashboard Page:

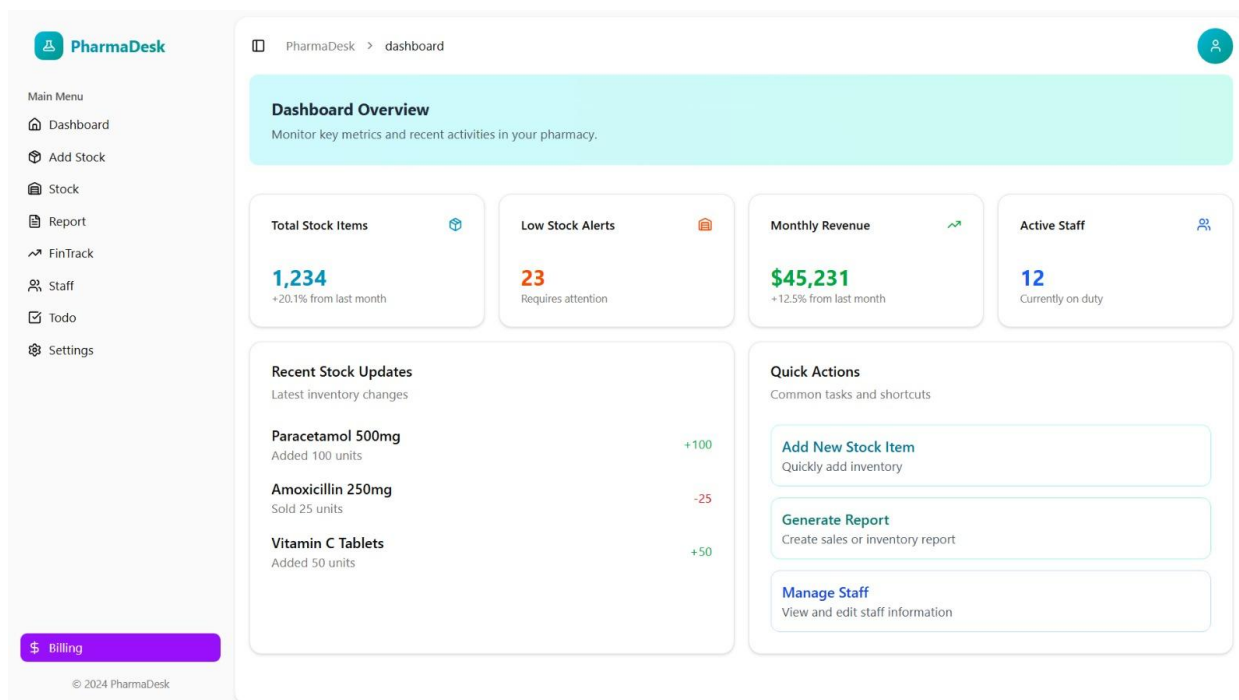


Figure 8.2 Dashboard Page Image

The image shows the Dashboard page of the PharmaDesk application, which provides a quick overview of key pharmacy operations. It displays important metrics such as total stock items, low stock alerts, monthly revenue, and active staff in summary cards. The dashboard also highlights recent stock updates and offers quick action options like adding new stock, generating reports, and managing staff, helping users monitor activities and take actions efficiently from a single screen.

8.3 Stock-Entry Page:

The screenshot shows the 'Stock Entry' form in PharmaDesk. The sidebar menu is identical to the dashboard. The main form area is titled 'Stock Entry' with the instruction 'Enter wholesaler and invoice information to update your inventory.' It contains a 'Wholesaler Details' section with fields for 'Wholesaler' (CureMed Pharma), 'Invoice No.' (CMP-003), and 'Date' (10-12-2025), along with 'Submit Entry' and 'Cancel' buttons. Below this is the 'Medicine Items' section, which shows '1 item added' and includes 'Upload Invoice' and 'Add Medicine' buttons. A summary bar indicates a 'Total Purchase Value' of ₹1540.00. The medicine entry details for 'Atorvastatin Lipicure' are as follows:

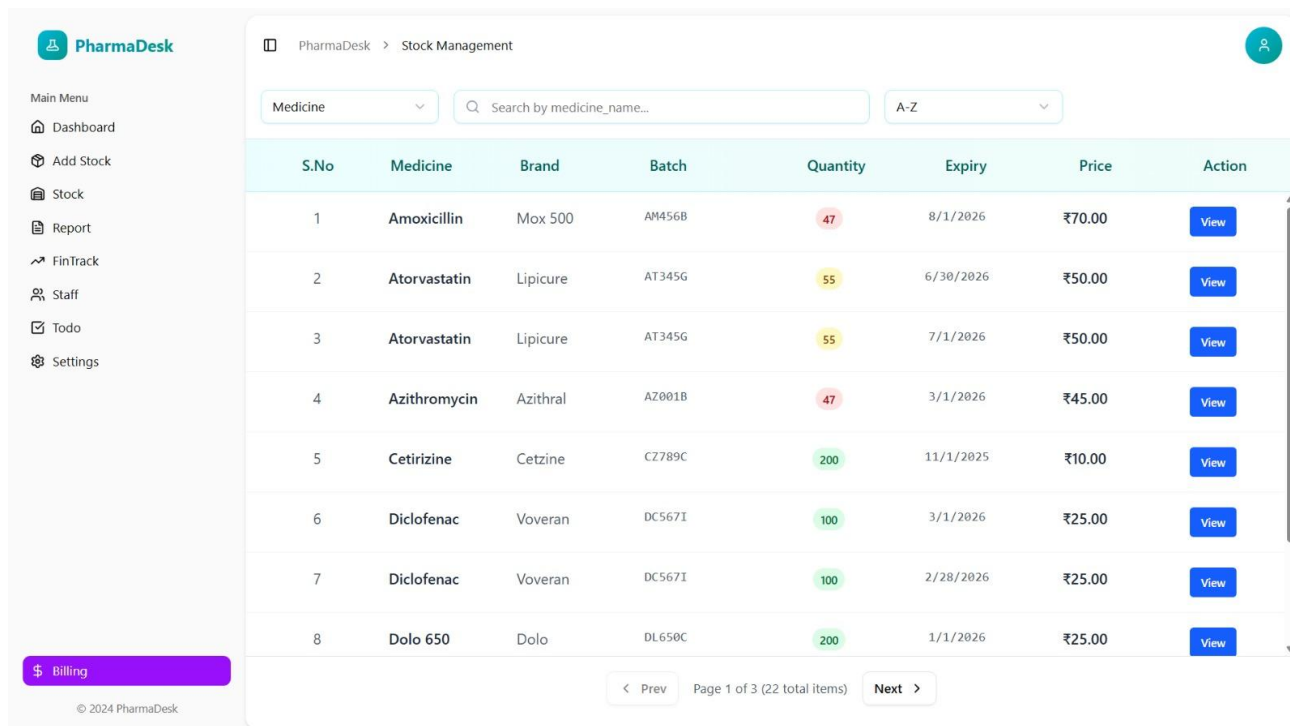
Batch No.	Quantity	MRP	Purchase
AT345G	55	₹50	₹28

Additional details for the medicine entry include 'Packed Type: Strip' and 'Expiry Date: 2026-06-30'.

Figure 8.3 Stock-Entry Page Image

The image shows the Stock Entry page of the PharmaDesk application, used to add new inventory into the system. It allows users to enter wholesaler details, including wholesaler name, invoice number, and date. Below this, medicine items can be added with details such as batch number, quantity, MRP, purchase price, manufacturing date, and expiry date. The page also displays the total purchase value and provides options to upload an invoice, add medicines, submit the entry, or cancel, making stock updating accurate and efficient

8.4 Stock-View Page



S.No	Medicine	Brand	Batch	Quantity	Expiry	Price	Action
1	Amoxicillin	Mox 500	AM456B	47	8/1/2026	₹70.00	View
2	Atorvastatin	Lipicure	AT345G	55	6/30/2026	₹50.00	View
3	Atorvastatin	Lipicure	AT345G	55	7/1/2026	₹50.00	View
4	Azithromycin	Azithral	AZ001B	47	3/1/2026	₹45.00	View
5	Cetirizine	Cetzine	CZ789C	200	11/1/2025	₹10.00	View
6	Diclofenac	Voveran	DC567I	100	3/1/2026	₹25.00	View
7	Diclofenac	Voveran	DC567I	100	2/28/2026	₹25.00	View
8	Dolo 650	Dolo	DL650C	200	1/1/2026	₹25.00	View

Figure 8.4 Stock-View Page Image

The image shows the Stock Management page of the Pharma Desk application, which provides a clear overview of available medicines in the pharmacy. It displays medicine details such as name, brand, batch number, quantity, expiry date, and price in a tabular format. A search and sorting option helps quickly locate medicines, while color-coded quantities indicate stock levels. The View button allows access to detailed medicine information, and pagination helps manage large inventories efficient

8.5 Report-Sales Page

S.No	Sale No	Employee	Total Amount	Payment	Action	Delete
1	00225DecIIN44	Neha Patel	₹150	UPI	View	Delete
2	00225DecIIN39	Neha Patel	₹250	Credit	View	Delete
3	00225DecIIMV9	Neha Patel	₹225	Cash	View	Delete
4	00225DecIIM0W	Neha Patel	₹360	Cash	View	Delete
5	00225DecIIHDM	Neha Patel	₹240	Cash	View	Delete
6	00225DecIIH8E	Neha Patel	₹160.5	Cash	View	Delete
7	00225DecHZZKP	Neha Patel	₹160.5	Cash	View	Delete

Page 1 of 1 (7 total items)

Figure 8.5 Report-Sales Page Image

The image shows the Sales List page of the PharmaDesk application, which provides an organized overview of all recorded sales transactions. It includes a search bar, sorting option, and an Export Excel button for easy data retrieval and reporting. The sales details are displayed in a table format showing sale number, employee name, total amount, and payment method. Each transaction can be viewed in detail or deleted, and pagination at the bottom helps navigate through records, making sales management simple and efficient.

8.6 Report-Sales Page-2

S.No	Sale No	Employee	Total Amount	Payment	Action	Delete
1	00225DecIIN44	Neha Patel	₹150	UPI	View	Delete
2	00225DecIIN39	Neha Patel	₹250	Credit	View	Delete
3	00225DecIIMV9	Neha Patel	₹225	Cash	View	Delete
4	00225DecIIM0W	Neha Patel	₹360	Cash	View	Delete
5	00225DecIIHDM	Neha Patel	₹240	Cash	View	Delete
6	00225DecIIH8E	Neha Patel	₹160.5	Cash	View	Delete
7	00225DecHZZKP	Neha Patel	₹160.5	Cash	View	Delete

Page 1 of 1 (7 total items)

Figure 8.6 Report-Sales Page-2 Image

AI Assistant for Pharmacist

The image shows a Sales List page of a pharmacy management system (Pharma Desk) with a Sale Details popup opened for a selected transaction. The popup displays complete information about a sale, including the sale number (00225DeclIN39), sale date and time, customer name and contact number, employee name, and payment method (Credit). On the left, it highlights the total bill amount of ₹250 along with the profit of ₹110. Below, an items list shows the medicine sold Atorvastatin, with its rate (₹50), quantity (5), and total (₹250). In the background, the Sales List table contains multiple sales entries with options to view or delete each record, along with features like search and export to Excel, making it easy to manage and review sales records efficiently.

8.7 AI Panel + Canvas

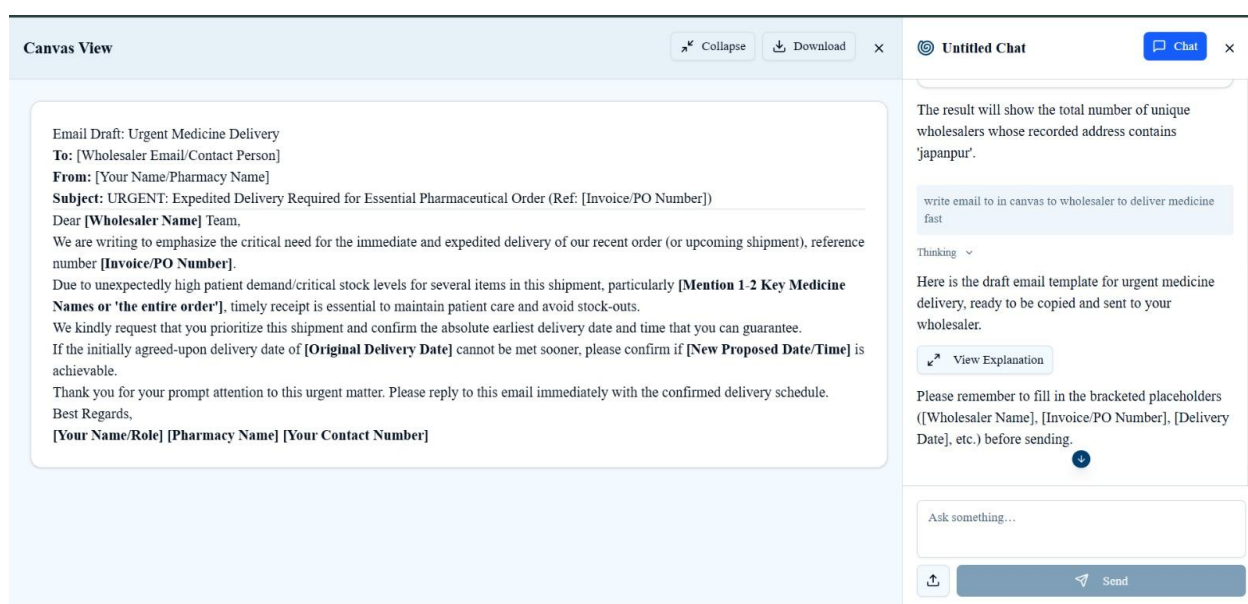


Figure 8.7 AI Panel + Canvas Page Image

The AI Assistant & Canvas Workspace module seamlessly integrates intelligent automation into the pharmacy workflow to simplify daily operations. The Interactive AI Panel on the right allows staff to use natural language commands to retrieve data, generate documents, or perform calculations, eliminating the need for manual searching or complex queries. On the left, the Dynamic Canvas View provides a clean, high-resolution space to display detailed outputs such as email drafts, official letters, or large data tables, ensuring clarity without cluttering the chat area. To enhance operational efficiency, the workspace includes options like Download and Collapse, enabling users to quickly export generated content for real-world use. Additionally, through contextual integration, the AI intelligently utilizes pharmacy-specific data—such as wholesaler details and invoice numbers—to automatically populate templates, saving time and reducing errors.

8.8 Billing-Page

POS Billing

Counter 1 ₹1080 Counter 2 Counter 3 Counter 4 Counter 5

Customer Details

Contact Number *
9876543210

Customer Name
Walk-in Customer

Payment Method
Cash

Cart Items
2 medicine(s) added

MEDICINE	BATCH	QTY	PRICE
Paracetamol	PC123A	12	₹240.00 @ 20
Amoxicillin	AM456B	12	₹840.00 @ 70

Total Items: 2

TOTAL PAYABLE
₹1080.00

Subtotal ₹1080.00
Tax (Incl.) ₹0.00
Net Total ₹1080.00

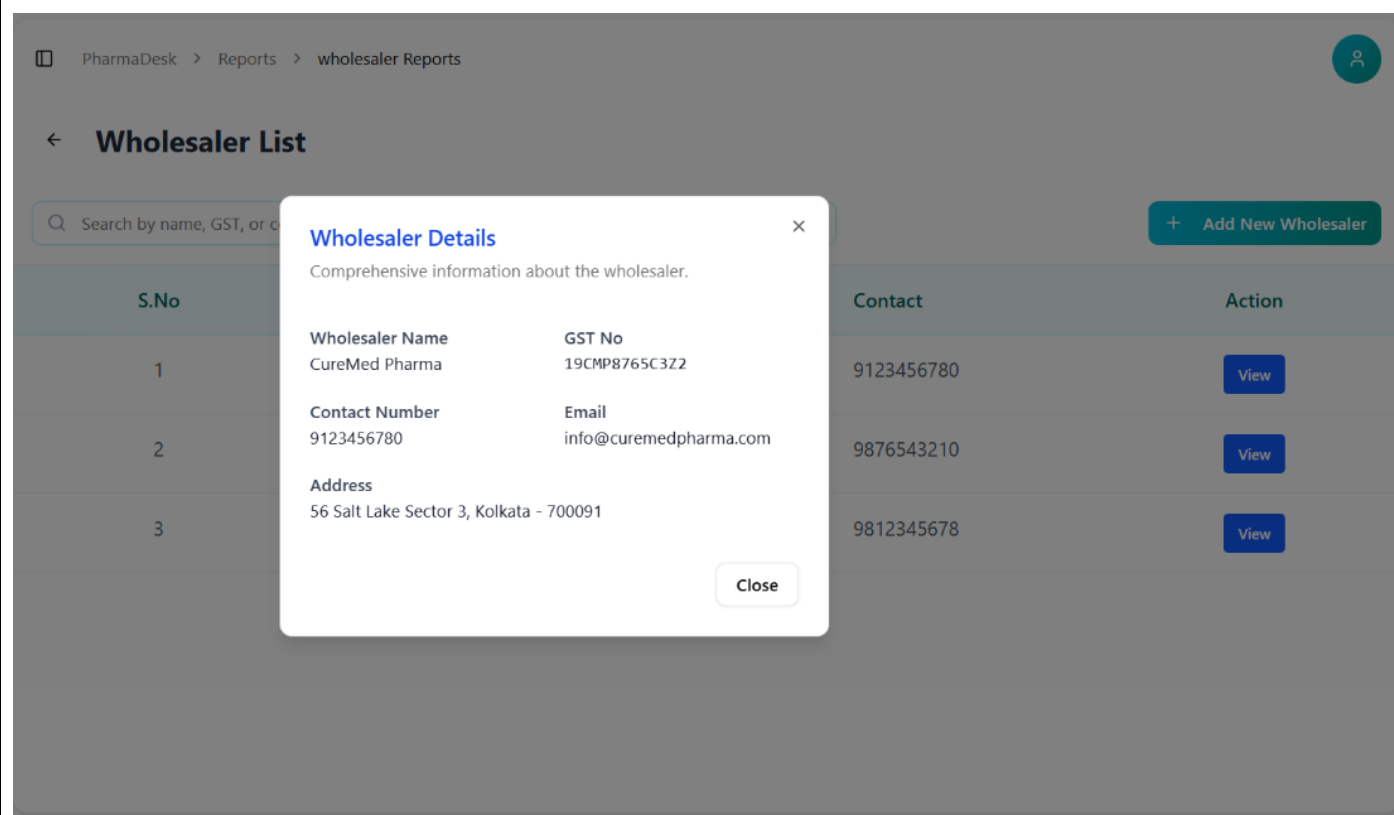
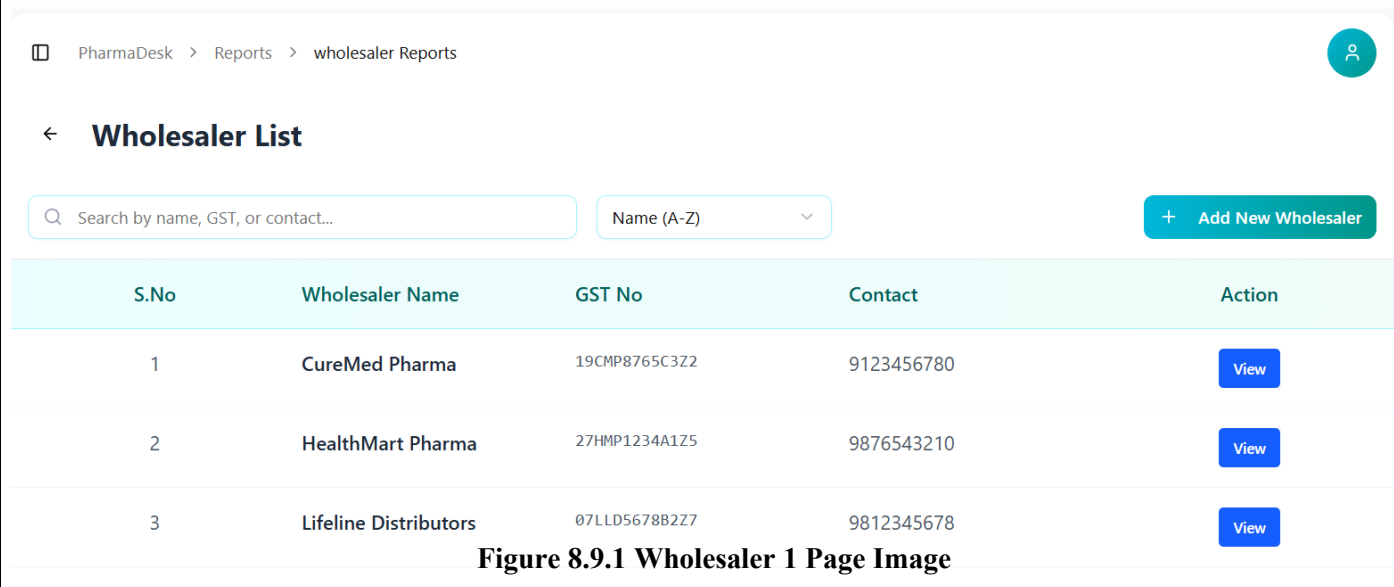
Generate Bill

Clear Counter 1

Figure 8.8 Billing Page Image

The Billing Page functions as a high-efficiency retail terminal designed to support fast and accurate customer checkout in a pharmacy environment. It includes a multi-session counter system that allows the operator to handle up to five billing sessions at the same time, making it easier to serve multiple customers without delays. The dynamic cart automatically manages medicines using batch IDs and quantities, instantly updating the subtotal and net total to avoid calculation errors. For better customer management, the page provides fields to enter customer names and contact numbers, enabling personalized billing and maintaining transaction history. To ensure financial clarity, a clearly visible Total Payable section displays the final amount, helping both the pharmacist and customer confirm the payment quickly and confidently.

8.9 Wholesaler-Page



The Wholesaler List page in the PharmaDesk application serves as a centralized directory for managing distributor relationships. The interface features a clean, tabular layout that displays essential business details such as Wholesaler Name, GST Number, and Contact Information. To enhance usability, the page includes a dynamic search bar for quick filtering and an "Add New Wholesaler" button for expanding the distributor database.

For more in-depth records, the system utilizes a "View" action that triggers a modal window, providing comprehensive details including the wholesaler's email and physical address without navigating away from the main list. This streamlined design ensures that administrators can efficiently monitor supply chain partners and access critical contact data through a single, intuitive dashboard.

8.10 Staff-Page

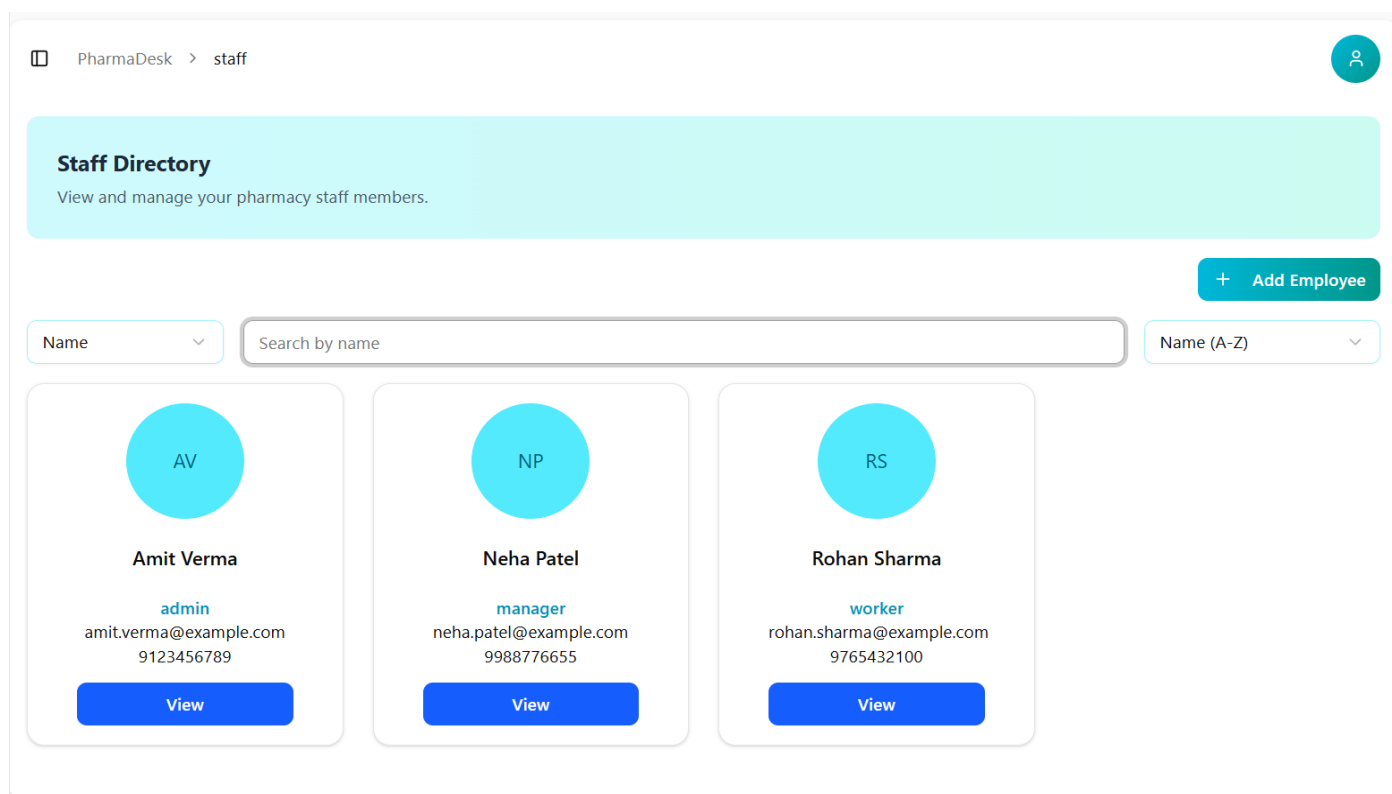


Figure 8.10 Staff Page Image

The **Staff Directory** page provides a visual overview of the pharmacy's workforce using a card-based layout. Each employee card displays key identification details, including their name, professional role (such as admin, manager, or worker), email address, and contact number. The interface includes an "**Add Employee**" action to expand the team and a search bar to filter staff by name.

8.11 Profile-Page-1

[PharmaDesk](#) > [Profile](#)



John Doe
Senior Pharmacist
john.doe@pharmaDesk.com

[Edit Profile](#)
[Change Password](#)

Personal Information
View your personal information

First Name
John

Last Name
Doe

Phone Number
+1 (555) 123-4567

Email Address
john.doe@pharmaDesk.com
Email cannot be changed

Date of Birth
15-05-1990

Role
Senior Pharmacist
Role cannot be changed

Address
123 Main Street, City, State 12345

Financial Information

Aadhar Number
1234 5678 9012

PAN Card
ABCDE1234F

Account Number
1234567890123456

Salary
\$75,000
Salary cannot be changed

Figure 8.11 Profile Page Image

The **Profile** page serves as a comprehensive personal dashboard for individual users, such as a Senior Pharmacist. It is divided into **Personal Information**, covering contact and address details, and **Financial Information**, which securely stores sensitive data like Aadhar, PAN, and salary details. Users can manage their account through "Edit Profile" and "Change Password" options, though certain fields like role and salary remain locked for administrative security.

CHAPTER 9

CONCLUSION

The development of this **Pharmacy Management System** represents a significant step forward in modernizing local pharmacy operations through the integration of Artificial Intelligence and robust software engineering. By bridging the gap between traditional inventory management and smart automation, the project successfully achieves the following:

- **Intelligent Automation:** The core innovation lies in the **AI-driven Assistant** and the **Semi-Agentive AI module**. By allowing pharmacists to add stock directly via image and PDF invoice analysis, the system eliminates manual data entry errors and drastically reduces the time required for procurement updates.
- **Data-Driven Decision Making:** The inclusion of a comprehensive **Dashboard** provides a bird's-eye view of the pharmacy's health, offering detailed summaries of sales, stock levels, and financial performance, enabling more informed business decisions.
- **Enterprise-Grade Security:** The system prioritizes data integrity and privacy through **JWT-based authentication, CORS policy implementation, and Role-Based Access Control (RBAC)**. By restricting specific AI functions for workers and ensuring secure communication, the application protects sensitive business information from unauthorized access.
- **Reliability and Accessibility:** With an **online backup system**, the pharmacy's data remains safe against local hardware failures. Furthermore, the system's design ensures high **portability**, allowing for easy installation on any PC and seamless usage on mobile devices within the same network.

Future Scope of the Project

- **voice-Activated Pharmacy Assistant:** Integrating a **Natural Language Processing (NLP) Voice Assistant** to allow for hands-free operation. Pharmacists could query stock availability ("Do we have Amoxicillin in Batch A?"), generate bills, or search for customer records using simple voice commands, significantly increasing speed during busy hours.
- **Agentic AI for Autonomous Stock Management:** Advancing the current AI into an "**Agentic**" **system** that doesn't just chat, but takes action. This AI agent would monitor the **medicine stock** table in real-time and, upon detecting low levels, automatically draft and send restock requests to wholesalers or update inventory records when a digital delivery note is received.
- **Cross-Platform Mobile Application:** Transitioning the project from a web-only platform to a **native mobile application (Android/iOS)**. This would allow the pharmacy owner to manage the store remotely and enable features like using the phone's camera as a high-speed barcode scanner for instant inventory logging and billing
- **E-Prescription & Doctor Portal:** Developing a secure API to link the pharmacy directly with **local clinics**. This would allow doctors to send digital prescriptions directly to the system's "Sales" queue, reducing manual data entry errors and preventing the risk of fraudulent or misinterpreted handwritten prescriptions.
- **Blockchain for Drug Traceability:** Implementing **Blockchain technology** to create an immutable audit trail for every medicine batch received from wholesalers. This would ensure 100% authenticity of the drugs sold, helping the pharmacy comply with global anti-counterfeit regulations and building higher customer trust

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